

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T29 · Sequences

With Answers

Questions with full worked solutions

■ 38 pages

FAMILY MAP

Sequences

F01 Recover arithmetic-series parameters

10 Q | 50 marks

F02 Use an arithmetic nth term

7 Q | 21 marks

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Worked Solution 15

Arithmetic series

UNIT 2

5 marks

20th term

$$a + 19d = 123.$$

Sum of 30 terms

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880 \Rightarrow 2a + 29d = 192.$$

Solve the pair

$$(2a + 38d) - (2a + 29d) = 246 - 192 \Rightarrow d = 6, \quad a = 9.$$

Final answer

$$a = 9, \quad d = 6$$

Question 01

4MA1-1H Q1

ORIGINAL QUESTION

3 marks

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

.....
(1)

Worked Solution 01

Arithmetic sequences

MODULAR UNIT 2

3 marks

General term

$$1 + 3(n - 1) = 3n - 2.$$

Twelfth term

$$5(12) + 17 = 77.$$

Final answer

$$3n - 2; \quad 77$$

Question 11

4MA1-1H Q24

ORIGINAL QUESTION

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

Worked Solution 11

Arithmetic sequence of polygon angles

MODULAR UNIT 2

6 marks

Two sums

$$S_n = \frac{n}{2}[168 + 4(n - 1)] = 2n^2 + 82n, \quad S_n = 180(n - 2).$$

Solve

$$n^2 - 49n + 180 = 0 \Rightarrow n = 45 \ (n > 5).$$

Angle sum

$$180(45 - 2) = 7740.$$

Final answer

$$7740^\circ$$

Question 15

4MA1-1H Q19 demand

ORIGINAL QUESTION UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

Worked solution 15

Last term of an arithmetic sequence

MODULAR UNIT 2

6 marks

Demand – Last term of an arithmetic sequenceIdentify a and d

$$a = 3, \quad d = 4.$$

Use the sum

$$S_n = \frac{n}{2}[2(3) + (n-1)4] = 7260 \Rightarrow 2n^2 + n - 7260 = 0.$$

Solve for the number of terms

$$(2n + 121)(n - 60) = 0 \Rightarrow n = 60.$$

Find the last term

$$x = 3 + (60 - 1)4 = 239.$$

Final answer

239

Question 16

4MA1-1H Q20

ORIGINAL QUESTION

6 marks

20 The sum of the first 80 terms of an arithmetic series, S , is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X =$

Worked Solution 16

Arithmetic-series reverse problem

MODULAR UNIT 2

6 marks

Use the two given facts

$$40(2a + 79d) = 470, \quad a + 74d = 14.5.$$

Solve for the sequence

$$2a + 79d = 11.75, \quad 2a + 148d = 29 \Rightarrow d = 0.25, \quad a = -4.$$

Use the partial sum

$$\frac{X}{2} [-8 + (X - 1)(0.25)] = 171.$$

Solve the quadratic

$$X^2 - 33X - 1368 = 0 = (X - 57)(X + 24).$$

Final answer

$$X = 57$$

Original question 12

4MA1-1H Q19 M01

MODULAR UNIT 2

5 marks

- 19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.

Show clear algebraic working.

.....

Worked solution 12

Find the sum of an arithmetic series

MODULAR UNIT 2

5 marks

M01 – Find the sum of an arithmetic series**Use the sum of 30 terms**

$$30a + 435d = 4395 \Rightarrow 2a + 29d = 293.$$

Use the 10th and 20th terms

$$(a + 9d) + (a + 19d) = 284 \Rightarrow 2a + 28d = 284.$$

Eliminate

$$d = 9, \quad a = 16.$$

Use the sum formula

$$S_{45} = \frac{45}{2} [2(16) + 44(9)] = 9630.$$

Final answer

9630

Original question 01

4MA1-1H Q1 Q1(a)

ORIGINAL QUESTION**2 marks**

- 1 Here are the first five terms of an arithmetic sequence.

1 5 9 13 17

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

Worked solution 01

Chapter: Sequences

MODULAR UNIT 2

2 marks

Q1(a) – Chapter: SequencesFamily: Use an arithmetic n th term • Variant: Arithmetic n th term from listed terms**Method**

The common difference is 4, so the n th term has the form $4n + c$.

Why: An arithmetic sequence with difference 4 has coefficient 4 on n .

Method

Using the first term, $4(1) + c = 1$, so $c = -3$ and the n th term is $4n - 3$.

Why: Substitute $n = 1$ to determine the constant term.

Final answer

n th term: $4n - 3$

Original question 02

4MA1-1H Q1 Q1(b)

ORIGINAL QUESTION**1 marks**

The n th term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m , for the $(2m)$ th term of this sequence.

.....
(1)

Worked solution 02

Chapter: Sequences

MODULAR UNIT 2

1 marks

Q1(b) – Chapter: SequencesFamily: Use an arithmetic n th term • Variant: Arithmetic term at a symbolic position**Method**Replace n by $2m$ in $3n + 5$: $3(2m) + 5 = 6m + 5$.*Why: The $(2m)$ th term is obtained by substituting $n = 2m$.***Final answer****term:** $6m + 5$

Original question 38

4MA1-2H Q26

ORIGINAL QUESTION**5 marks**

- 26** An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

Worked solution

Family: Recover arithmetic-series parameters • Variant: Nonlinear or integer-constrained series recovery

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 38 MODULAR UNIT 2
Chapter: Sequences 5 marks
Q26 – Chapter: Sequences
Family: Recover arithmetic-series parameters • Variant: Nonlinear or integer-constrained series recovery
Method
$S_m = m/2[2a + (m - 1)d] = 39$ and $S_{2m} = m[2a + (2m - 1)d] = 320$.
Why: Use the arithmetic-series sum formula twice.
Method
Subtracting the equations eliminates a and gives $dm = 242$.
Why: Compare the two sum equations.
Method
$242 = 2 \times 121$; since d is prime, $d=2$ and $m = 121$.
Why: Use the prime restriction on d.
Method
$m=11$ because m is positive.
Why: Take the positive square root.
Method
Thus $d=2$ and $m=11$.
Why: State both requested values.
Final answer
d,m: $d=2$, $m=11$

FINAL ANSWER

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$k = \dots\dots\dots$

Worked solution

January 2020 | Question 25 | 3 marks

Family: Recover arithmetic-series parameters • Variant: Recover first term and common difference

Method / working

1. There are 50 terms from 2021 to 2070.
2. Arithmetic sum: $33125 = 50/2[2(50) + 49k]$.
3. $1325 = 100 + 49k$, so $k = 25$.

Final answer: $k = 25$

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .

Show your working clearly.

$a = \dots\dots\dots$

Worked solution

January 2021 | Question 24 | 4 marks

Family: Recover arithmetic-series parameters | Variant: Symbolic relation between partial sums

METHOD / WORKING

1. Write $S_n = n[2a + (2n-1)d]$.
2. Set $S_n = 4S_n$ and cancel n .
3. Simplify to $2a - d = 4a - 2d$.
4. Therefore $a = d/2$.

FINAL ANSWER

$$a = d/2$$

2 Here are the first 4 terms of an arithmetic sequence.

85 79 73 67

Find an expression, in terms of n , for the n th term of the sequence.

Worked solution

June 2021 | Question 2 | 2 marks

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

Method / working

1. The common difference is -6, so use $a + (n-1)d$.
2. $85 + (n-1)(-6)$ simplifies to $91 - 6n$.

Final answer: $91 - 6n$

- 23** The sum of the first N terms of an arithmetic series, S , is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N .

Show clear algebraic working.

$N = \dots\dots\dots$

Worked solution

June 2021 | Question 23 | 5 marks

Family: Recover arithmetic-series parameters | Variant: Recover number of arithmetic terms

METHOD / WORKING

1. From the 2nd and 5th terms, solve $a+d=8.5$ and $a+4d=13$ to get $a=7, d=1.5$.
2. *Substitute into* $S_N = N/2[2a + (N - 1)d] = 292$.
3. *Rearrange* to the quadratic $3N^2+25N-1168=0$.
4. Factor or use the quadratic formula to obtain $N=16$ (discard the negative root).
5. Check $S_{16}=292$ and state $N=16$.

FINAL ANSWER

N = 16

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

Worked solution

November 2021 | Question 20 | 5 marks

Family: Recover arithmetic-series parameters | Variant: Recover first term and common difference

METHOD / WORKING

1. The first four terms are
2. $k, \frac{3k}{4}, \frac{k}{2}, \frac{k}{4}$
3. The common difference is
4. $d = -\frac{k}{4}$
5. The 15th term is
6. $u_{15} = k + 14(-\frac{k}{4})$
7. $u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$
8. Given
9. $-\frac{5k}{2} = 90 + 2k$
10. $-5k = 180 + 4k$
11. $k = -20$
12. So
13. $a = -20, d = 5$
14. $S_{30} = \frac{30}{2}(2(-20) + 29(5))$
15. $S_{30} = 15(105) = 1575$
16. Answer: 1575

FINAL ANSWER

1575

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

Answer reference | May-Nov 2020 | Question 24

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

24	$a = 8 \quad d = 7$		4	M1 can be implied
	$(S_{100} =) \frac{100}{2} (2 \times 8 + (100 - 1) \times 7) (= 35\,450) \text{ or}$ $(S_{49} =) \frac{49}{2} (2 \times 8 + (49 - 1) \times 7) (= 8624) \text{ or}$ $(S_{50} =) \frac{50}{2} (2 \times 8 + (50 - 1) \times 7) (= 8975)$			M1
	'35450' - '8624' or '35450' - '8975' + $(8 + (50 - 1) \times 7)$			M1
		26 826		A1
				Total 4 marks
	Alternative scheme			
	$(u_n =) 7n + 1$	$a = 8 \text{ and } d = 7$	4	M1 can be implied
	$(u_{50} =) 7 \times 50 + 1 (= 351) \text{ or}$ $(u_{100} =) 7 \times 100 + 1 (= 701)$	$(u_{50} =) 8 + (50 - 1) \times 7$ $(= 351)$		M1
	$\frac{51}{2} ('351' + '701')$	$\frac{51}{2} (2 \times 351 + (51 - 1) \times 7)$		M1
		26 826		A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 24 | 4 marks

Family: Use an arithmetic nth term | Variant: Arithmetic nth term from listed terms

METHOD / WORKING

Worked solution
May-Nov 2020 Question 24 4 marks
Family: Use an arithmetic nth term Variant: Arithmetic nth term from listed terms
Method
The arithmetic sequence is
$8, 15, 22, 29, 36, \dots$
So
$a=8, d=7$
The nth term is
$u_n = 8 + 7(n - 1) = 7n + 1$
$u_{(50)} = 7(50) + 1 = 351$
$u_{(100)} = 7(100) + 1 = 701$
There are
$100 - 50 + 1 = 51$
terms from the 50th to the 100th inclusive.
$\text{sum} = (51)(2)(351 + 701)$
$= 26826$
Answer: 26826

FINAL ANSWER

- 22 The first term of an arithmetic series is $(2t + 1)$ where $t > 0$
The n th term of this arithmetic series is $(14t - 5)$

The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.



$p =$ $q =$ $r =$

(Total for Question 22 is 4 marks)

Answer reference | January 2023 | Question 22

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

22	$2t + 1 + (n - 1)3 = 14t - 5$		4	M1 for the use of n th term $= a + (n - 1)d$ to find n
	$(n =) 4t - 1$ or $(n =) -1 + 4t$			A1
	$(S_n =) \frac{"4t-1"}{2} [2(2t+1) + ("4t-1"-1)3]$ or $(S_n =) \frac{"4t-1"}{2} [2t+1+14t-5]$ oe			M1 for the use of S_n formula (must be in terms of t) Allow their expression for n dep on M1
	Working required	$p = 2$ $q = 4$ $r = 2$		A1 dep on M2 allow $(S_n =) 2(4t-1)^2$ Values of p , q and r must come from correct working
				Total 4 marks

Worked solution

January 2023 | Question 22 | 4 marks

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

Official final mark-scheme pages are embedded immediately after this question.

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86
- The sum of the first N terms of the series is 5194
- Work out the value of N
Show your working clearly.

$N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

Worked solution

November 2025 | Question 25 | 5 marks

Family: Use an arithmetic nth term | Variant: Arithmetic nth term from listed terms

METHOD / WORKING

Worked solution
November 2025 Question 25 5 marks
Family: Use an arithmetic nth term Variant: Arithmetic nth term from listed terms
Method
$a=10$
$u_{(20)}=86$
$10+19d=86$
$d=4$
Use
$S_N = \frac{N}{2}(2a + (N-1)d)$
$5194 = \frac{N}{2}(20+4(N-1))$
$5194 = \frac{N}{2}(4N+16)$
$5194 = 2N(N+4)$
$N^2 + 4N - 2597 = 0$
$N = \frac{-4 \pm \sqrt{16 + 4 \times 2597}}{2} = 49$
Answer: $N=49$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER